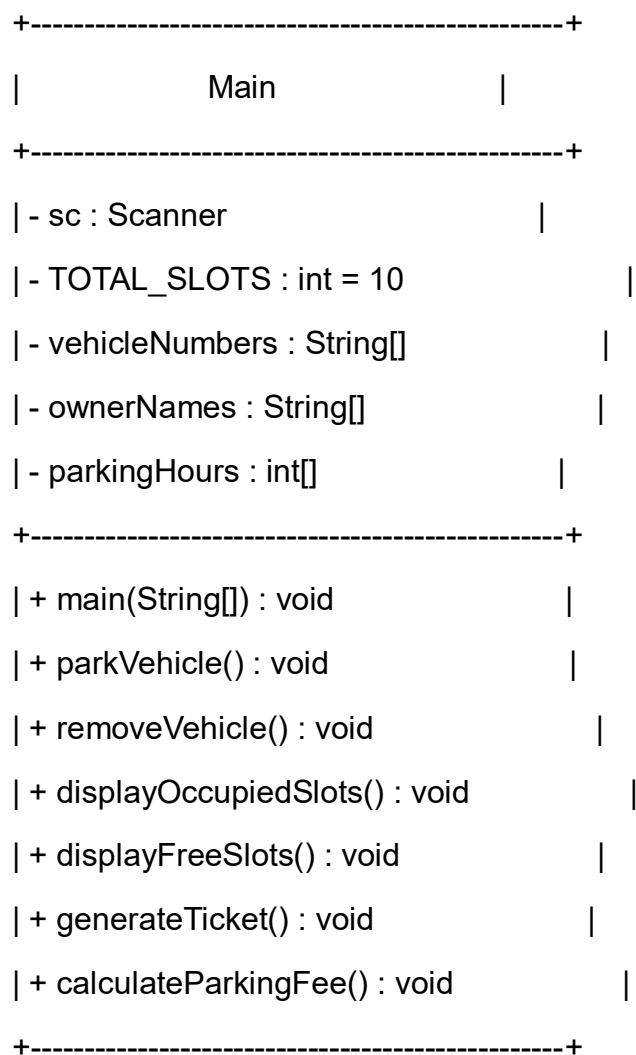


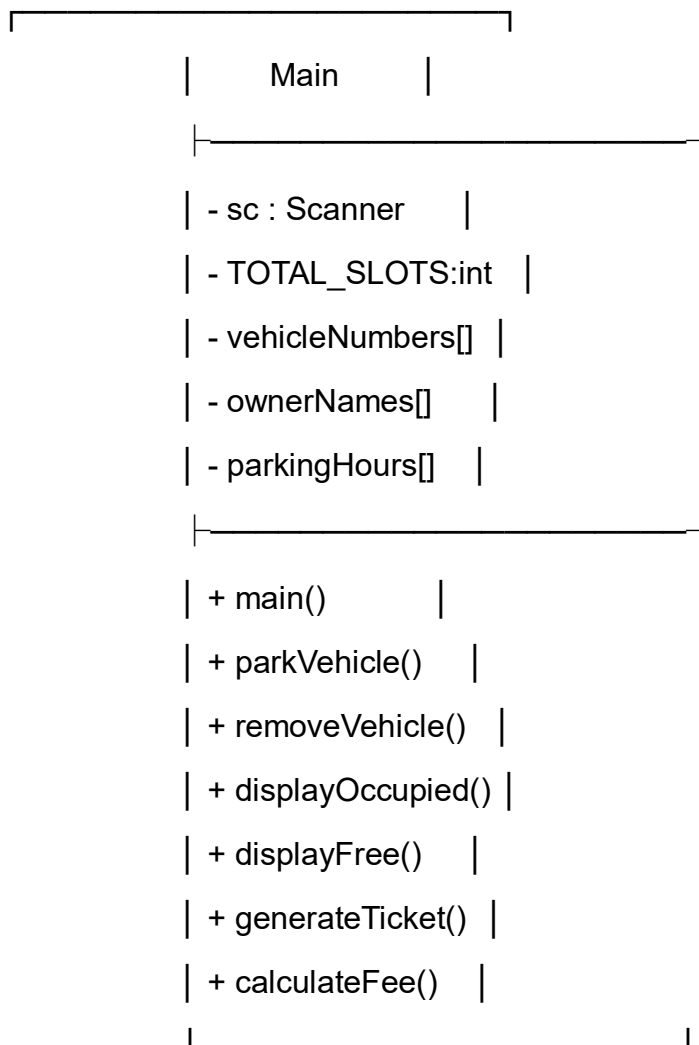
Parking lot system with SOLID principles in java

- s1. Park a vehicle
- s2. Remove a vehicle
- s3. Display free slot
- s4. Display occupied slot
- s5. Generate ticket
- s6. Calculate parking fee -> weekdays=3 hrs 30rupees, weekends= 3hrs 40 rupees

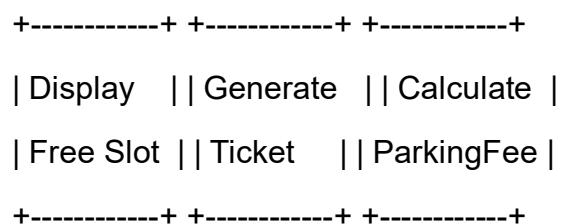
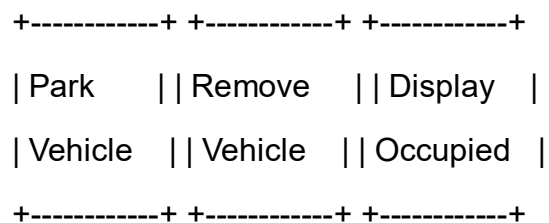
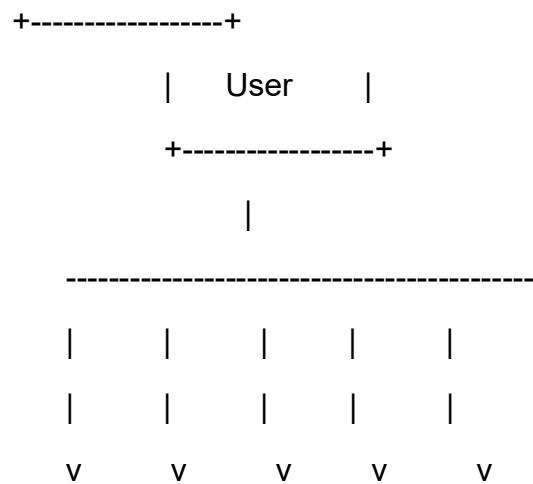
Unified Modeling Language (UML) Class diagram



Unified Modeling Language (UML) Class diagram (Visual)



Unified Modeling Language (UML) Use case diagram



Viva Explanation

Class Name : Main

Attributes:

- Scanner object for input
- Total parking slots
- Vehicle number array
- Owner name array
- Parking duration array

Methods:

- parkVehicle()
- removeVehicle()
- displayOccupiedSlots()
- displayFreeSlots()
- generateTicket()
- calculateParkingFee()

Actor:

- User

Use Cases:

- Park Vehicle
- Remove Vehicle
- Display Occupied Slots
- Display Free Slots
- Generate Ticket
- Calculate Parking Fee

Code

```
import java.util.Scanner;
```

```

public class Main{

    static Scanner sc = new Scanner(System.in);

    static final int TOTAL_SLOTS = 10;

    static String[] vehicleNumbers = new String[TOTAL_SLOTS];
    static String[] ownerNames = new String[TOTAL_SLOTS];
    static int[] parkingHours = new int[TOTAL_SLOTS];

    public static void main(String[] args) {

        while (true) {

            System.out.println("\n=====");
            System.out.println("    PARKING LOT SYSTEM");
            System.out.println("=====");
            System.out.println("1. Park Vehicle");
            System.out.println("2. Remove Vehicle");
            System.out.println("3. Display Occupied Slots");
            System.out.println("4. Display Free Slots");
            System.out.println("5. Generate Ticket");
            System.out.println("6. Calculate Parking Fee");
            System.out.println("7. Exit");
            System.out.println("=====");

            System.out.print("Enter Choice : ");
            int choice = sc.nextInt();
            sc.nextLine();

```

```
switch (choice) {  
  
    case 1:  
        parkVehicle();  
        break;  
  
    case 2:  
        removeVehicle();  
        break;  
  
    case 3:  
        displayOccupiedSlots();  
        break;  
  
    case 4:  
        displayFreeSlots();  
        break;  
  
    case 5:  
        generateTicket();  
        break;  
  
    case 6:  
        calculateParkingFee();  
        break;  
  
    case 7:  
        System.out.println("\nThank You...");
```

```
System.exit(0);
```

```
default:
```

```
System.out.println("Invalid Choice!");
```

```
}
```

```
}
```

```
}
```

```
static void parkVehicle() {
```

```
System.out.println("\n=====");
```

```
System.out.println("      PARK VEHICLE");
```

```
System.out.println("=====");
```

```
int slot = -1;
```

```
for (int i = 0; i < TOTAL_SLOTS; i++) {
```

```
    if (vehicleNumbers[i] == null) {
```

```
        slot = i;
```

```
        break;
```

```
    }
```

```
}
```

```
if (slot == -1) {
```

```
    System.out.println("Parking Lot Full!");
```

```
    return;
```

```
}
```

```
System.out.print("Enter Vehicle Number : ");
```

```
String vehicleNo = sc.nextLine();
```

```
for (int i = 0; i < TOTAL_SLOTS; i++) {
```

```
    if (vehicleNumbers[i] != null &&  
        vehicleNumbers[i].equalsIgnoreCase(vehicleNo)) {
```

```
        System.out.println("Vehicle Already Parked!");
```

```
        return;
```

```
    }
```

```
}
```

```
System.out.print("Enter Owner Name : ");
```

```
String owner = sc.nextLine();
```

```
System.out.print("How Many Hours Do You Want To Park ? : ");
```

```
int hours = sc.nextInt();
```

```
vehicleNumbers[slot] = vehicleNo;
```

```
ownerNames[slot] = owner;
```

```
parkingHours[slot] = hours;
```

```
System.out.println("\nVehicle Parked Successfully");
```

```
System.out.println("Allocated Slot : " + (slot + 1));
```

```
System.out.println("Parking Duration : " + hours + " Hours");
```

```
}
```

```
static void removeVehicle() {
```



```

System.out.println("\n=====");
System.out.println("    REMOVE VEHICLE");
System.out.println("=====");

System.out.print("Enter Vehicle Number : ");
String vehicleNo = sc.nextLine();

for (int i = 0; i < TOTAL_SLOTS; i++) {

    if (vehicleNumbers[i] != null &&
        vehicleNumbers[i].equalsIgnoreCase(vehicleNo)) {

        vehicleNumbers[i] = null;
        ownerNames[i] = null;
        parkingHours[i] = 0;

        System.out.println("Vehicle Removed Successfully");
        return;
    }
}

System.out.println("Vehicle Not Found!");
}

static void displayOccupiedSlots() {

    System.out.println("\n=====");
    System.out.println("    OCCUPIED SLOTS");
    System.out.println("=====");

```

```

boolean found = false;

for (int i = 0; i < TOTAL_SLOTS; i++) {

    if (vehicleNumbers[i] != null) {

        System.out.println("Slot Number    : " + (i + 1));
        System.out.println("Vehicle Number : " + vehicleNumbers[i]);
        System.out.println("Owner Name    : " + ownerNames[i]);
        System.out.println("Parking Duration : " + parkingHours[i] + " Hours");
        System.out.println("-----");

        found = true;
    }
}

if (!found) {
    System.out.println("No Occupied Slots");
}

static void displayFreeSlots() {

    System.out.println("\n=====");
    System.out.println("    FREE SLOTS");
    System.out.println("=====");

    boolean found = false;

```

```

for (int i = 0; i < TOTAL_SLOTS; i++) {

    if (vehicleNumbers[i] == null) {

        System.out.println("Slot " + (i + 1) + " is Free");
        found = true;
    }
}

if (!found) {
    System.out.println("No Free Slots Available");
}
}

static void generateTicket() {

    System.out.println("\n=====");
    System.out.println("    PARKING TICKET");
    System.out.println("=====");

    System.out.print("Enter Vehicle Number : ");
    String vehicleNo = sc.nextLine();

    for (int i = 0; i < TOTAL_SLOTS; i++) {

        if (vehicleNumbers[i] != null &&
            vehicleNumbers[i].equalsIgnoreCase(vehicleNo)) {

```

```

        System.out.println("Vehicle Number   : " + vehicleNumbers[i]);
        System.out.println("Owner Name      : " + ownerNames[i]);
        System.out.println("Slot Number    : " + (i + 1));
        System.out.println("Parking Duration : " + parkingHours[i] + " Hours");

        return;
    }
}

System.out.println("Vehicle Not Found!");
}

```

```

static void calculateParkingFee() {

    System.out.println("\n=====");
    System.out.println("      PARKING BILL");
    System.out.println("=====");

    System.out.print("Enter Vehicle Number : ");
    String vehicleNo = sc.nextLine();

    for (int i = 0; i < TOTAL_SLOTS; i++) {

        if (vehicleNumbers[i] != null &&
            vehicleNumbers[i].equalsIgnoreCase(vehicleNo)) {

            int hours = parkingHours[i];

            int blocks = (int) Math.ceil(hours / 3.0);

```

```
double fee = blocks * 40;
```

```
System.out.println("Vehicle Number : " + vehicleNumbers[i]);
```

```
System.out.println("Owner Name    : " + ownerNames[i]);
```

```
System.out.println("Slot Number   : " + (i + 1));
```

```
System.out.println("Duration      : " + hours + " Hours");
```

```
System.out.println();
```

```
System.out.println("Rate          : Rs.40 Per 3 Hours");
```

```
System.out.println("Amount         : Rs." + fee);
```

```
System.out.println("=====");
```

```
return;
```

```
}
```

```
}
```

```
System.out.println("Vehicle Not Found!");
```

```
}
```

```
}
```